IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

IBM Applied Data Science Capstone
Reconstructed course-data analysis
27 October 2026



Outline

Executive summary and project question

Methodology and data provenance

EDA charts and SQL results

Folium maps and Plotly Dash views

Classification results and conclusions

Appendix: sources and reproducibility

Completed notebooks and Python files:
https://github.com/mohitpingale24-alt/IBM-Stock-Data-Analysis/tree/main/spacex-capstone

Executive Summary

60 of 90 launches have Class 1 outcomes (66.7%). The course label includes ocean landings.

KSC LC 39A has 17/22 Class 1 outcomes in the EDA snapshot. Site differences also reflect mission mix and time.

The decision tree scores 83.3% on 18 test launches. KNN leads training cross-validation by a small margin.

Historical associations support further analysis, not current launch or price predictions.

Introduction

Question: Which recorded launch characteristics are associated with Falcon 9 first-stage landing outcomes?

Reusable first stages motivate the analysis, but these datasets contain no recovery cost or bid-price target.

We compare launch site, payload, orbit and flight history, then evaluate four classifiers.

Scope: historical course snapshots, with new calculations and reconstructed map/dashboard views.

Section
1

Methodology



Methodology

1 Obtain and validate the published IBM course snapshots

2 Audit missing values, outcome labels and dataset boundaries

3 Execute SQLite queries and explore launch characteristics

4 Rebuild Folium maps and the interactive Dash views

5 Split the modeling data, tune within training folds and evaluate once on a held-out test set

Data Collection

Source	Records	Use in this report
dataset_part_1 / part_2	90 each	Wrangling, EDA and ML
Spacex.csv	101	SQL tasks
spacex_launch_geo.csv	56	Folium mapping
spacex_launch_dash.csv	56	Dash interactions

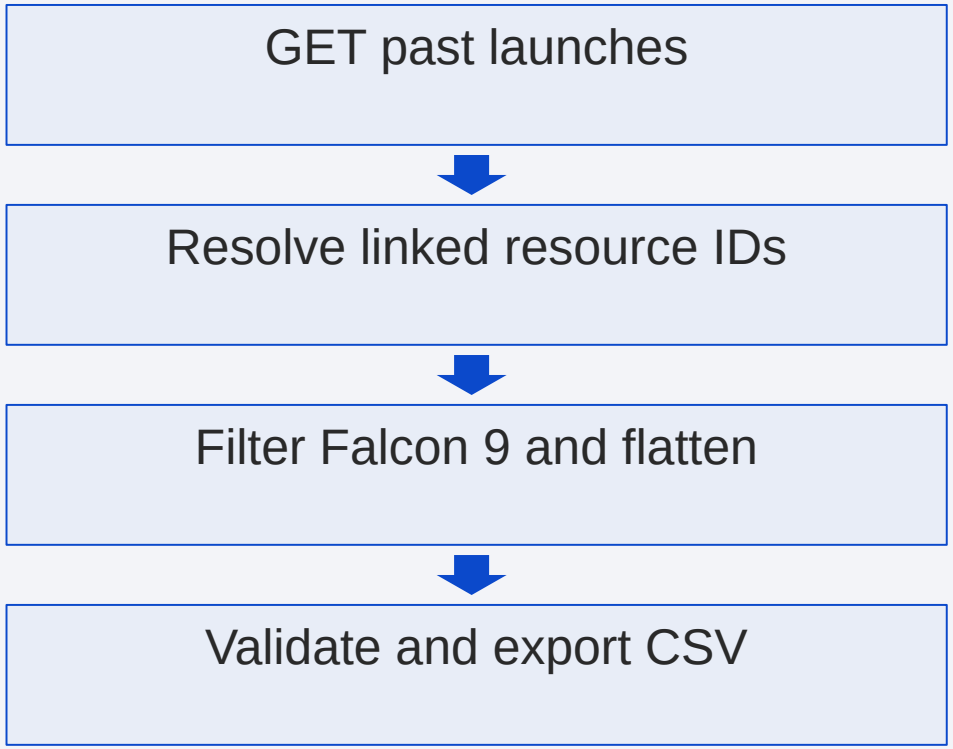
Keep snapshots separate. Different row counts and labels prevent direct pooling.

Data Collection – SpaceX API

Reconstructed API workflow

Resolve nested identifiers to obtain rocket, launchpad, payload and core attributes. Keep Falcon 9 launches and flatten the fields.

Results here use IBM’s published 90-row CSV. A fresh API harvest was not performed.

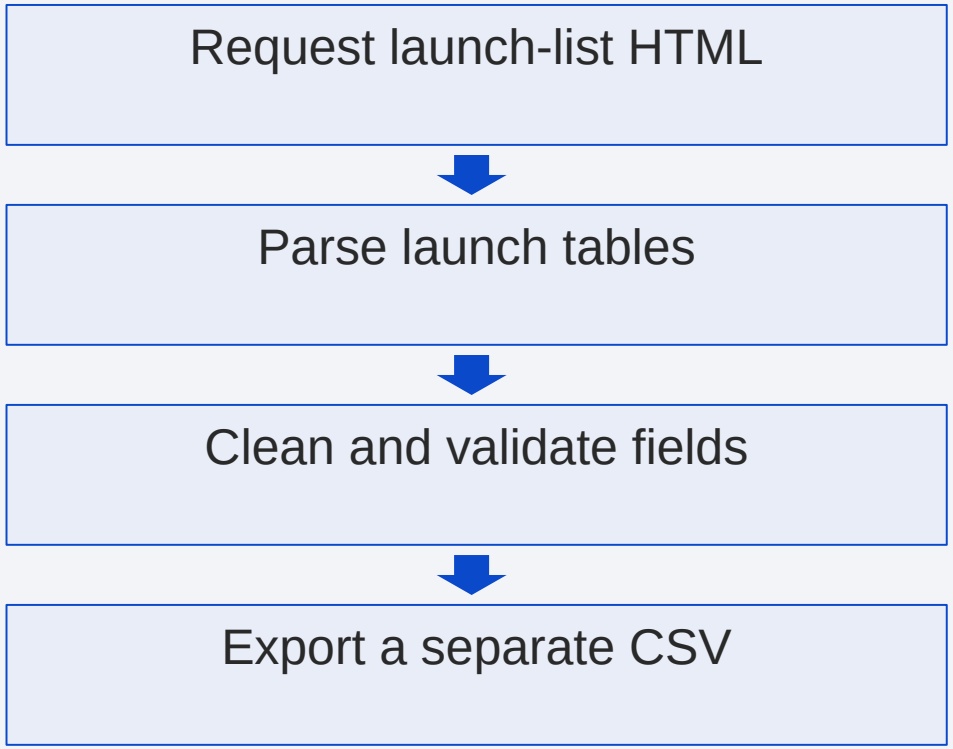


Data Collection – Web Scraping

Reconstructed scraping workflow

Extract date, booster, site, payload, customer and outcomes. Normalize mass units and remove footnote markers.

Results here use the course SQL snapshot. An independent Wikipedia scrape is now included in the repository. It does not replace the modeling snapshot.



Data Wrangling

Audit: 90 unique flight numbers, dated 4 June 2010 to 5 November 2020.

The published payload column already contains imputed values. LandingPad has 26 missing entries and is excluded from modeling.

Class 1 = True ASDS, True RTLS or True Ocean. Class 0 combines failure, unknown and no-landing outcomes.

Exclude Outcome from predictors. Fit numeric scaling and categorical encoding inside training folds.

EDA with Data Visualization

Flight number versus site and orbit: inspect changes across the launch sequence.

Payload versus site and orbit: compare mission profiles and outcome overlap.

Orbit success bars: compare Class 1 proportions with group sizes.

Yearly success line: inspect historical change. No 2011 observation exists in this snapshot.

All six charts use the same 90-row dataset.

EDA with SQL

Executed ten SQLite queries on 101 course records.

DISTINCT and LIKE identify launch sites and CCA records.
SUM and AVG summarize payload mass.
MIN identifies the first recorded successful ground landing.
Filters identify drone-ship landings and 2015 failures.
GROUP BY counts mission and landing outcomes.
A MAX subquery returns all tied maximum-payload boosters.

Normalize trailing spaces when interpreting mission outcomes.

Build an Interactive Map with Folium

Circle markers identify the four site labels in the course file.

A marker cluster contains all 56 launch outcomes. Green means Class 1 and red means Class 0.

Mouse coordinates help inspect the area. Lines show straight-line distances to selected proximity points.

The proximity example uses approximate sample points, not surveyed nearest-feature distances. Course coordinates also require validation before operational use.

Build a Dashboard with Plotly Dash

The site selector switches between all launches and one site.

For all sites, the pie shows each site’s share of successful outcomes. For one site, it shows the Class 0 / Class 1 ratio.

The payload slider filters the scatter plot. Booster version supplies the color grouping.

Screenshots show a locally reconstructed Dash application using the official 56-row course file.

Predictive Analysis (Classification)

Stratified 80/20 split, seed 42: 72 training and 18 test launches.

Five-fold stratified cross-validation tunes logistic regression, SVM, decision tree and KNN on training data only.

Numeric pipeline: median imputation and standard scaling. Categorical pipeline: one-hot encoding with unseen categories ignored.

Choose by mean validation accuracy. Report test accuracy and a confusion matrix without retuning on the test set.

Results

EDA: 66.7% Class 1 overall, with substantial variation across orbit and launch year.

SQL: NASA (CRS) payload totals 45,596 kg. The first recorded ground-pad success is 22 December 2015.

Dash: KSC LC-39A contributes 10 of 24 successful outcomes and has the highest site ratio, 10/13.

ML: validation-selected KNN scores 14/18 on the test set. The decision tree scores 15/18.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Later flights contain more Class 1 outcomes.

Site rates: CCAFS 60.0%, KSC 77.3%, VAFB 76.9%.

1 CCAFS SLC 40
2 KSC LC 39A
3 VAFB SLC 4E



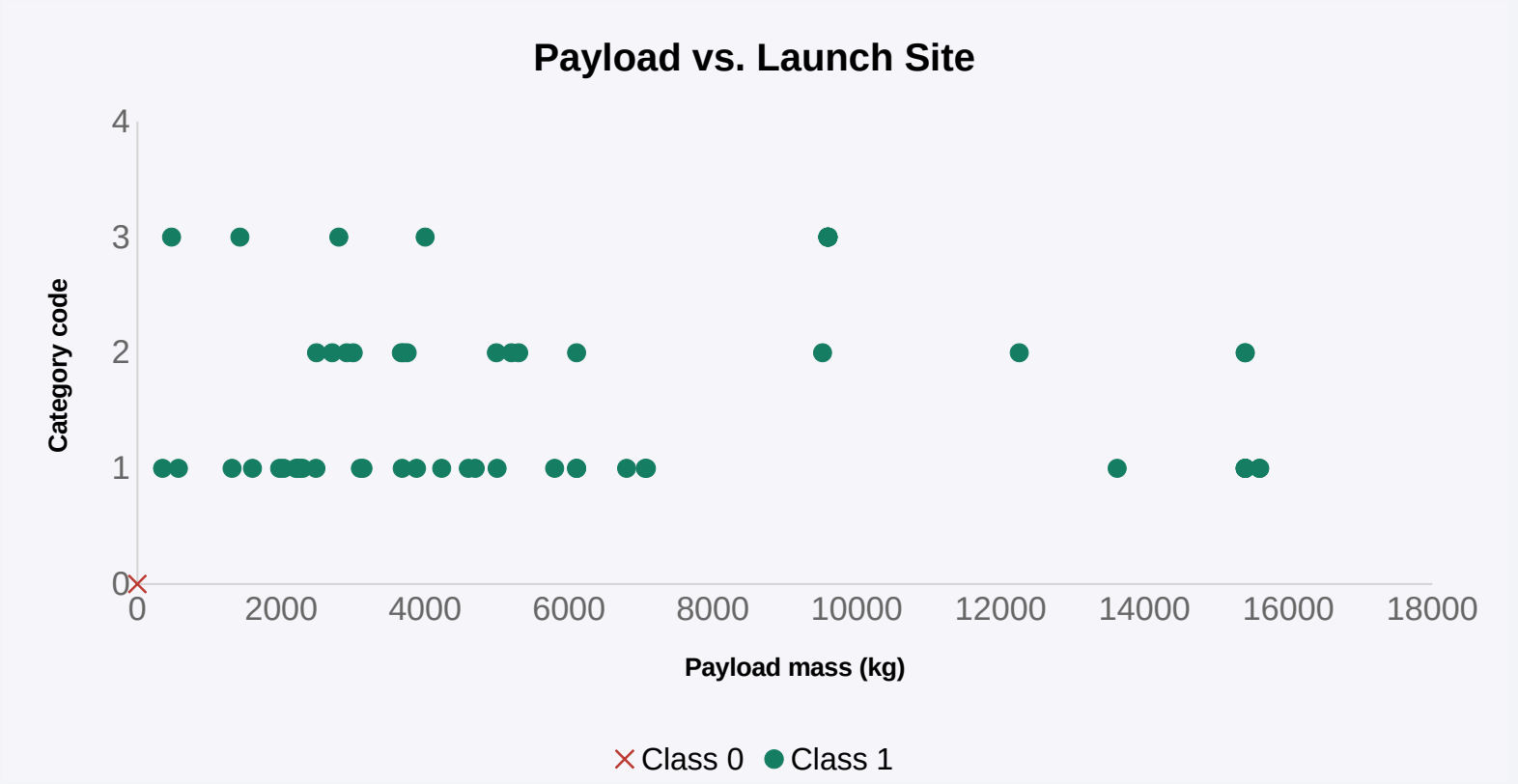
IBM EDA snapshot: 90 launches. Class 1 includes five successful ocean outcomes.

Payload vs. Launch Site

Payload ranges overlap across sites.

Payload alone does not separate Class 0 and Class 1. Site comparisons also reflect orbit and launch year.

1 CCAFS SLC 40
2 KSC LC 39A
3 VAFB SLC 4E



IBM EDA snapshot: 90 launches. Class 1 includes five successful ocean outcomes.

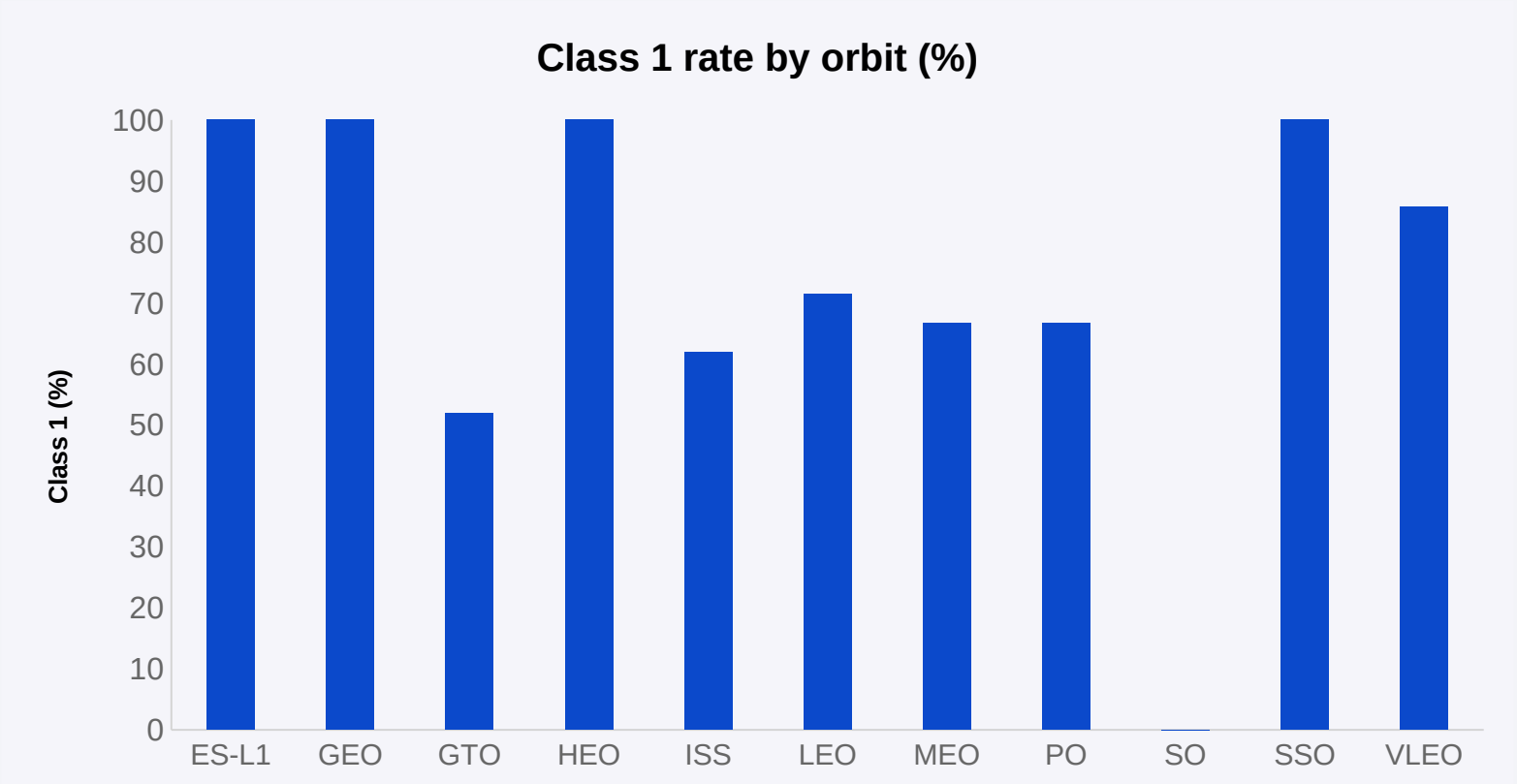
Success Rate vs. Orbit Type

GTO: 14/27 (51.9%).

VLEO: 12/14 (85.7%).

ES-L1, GEO and HEO each contain one launch.

SSO: 5/5 (100%).

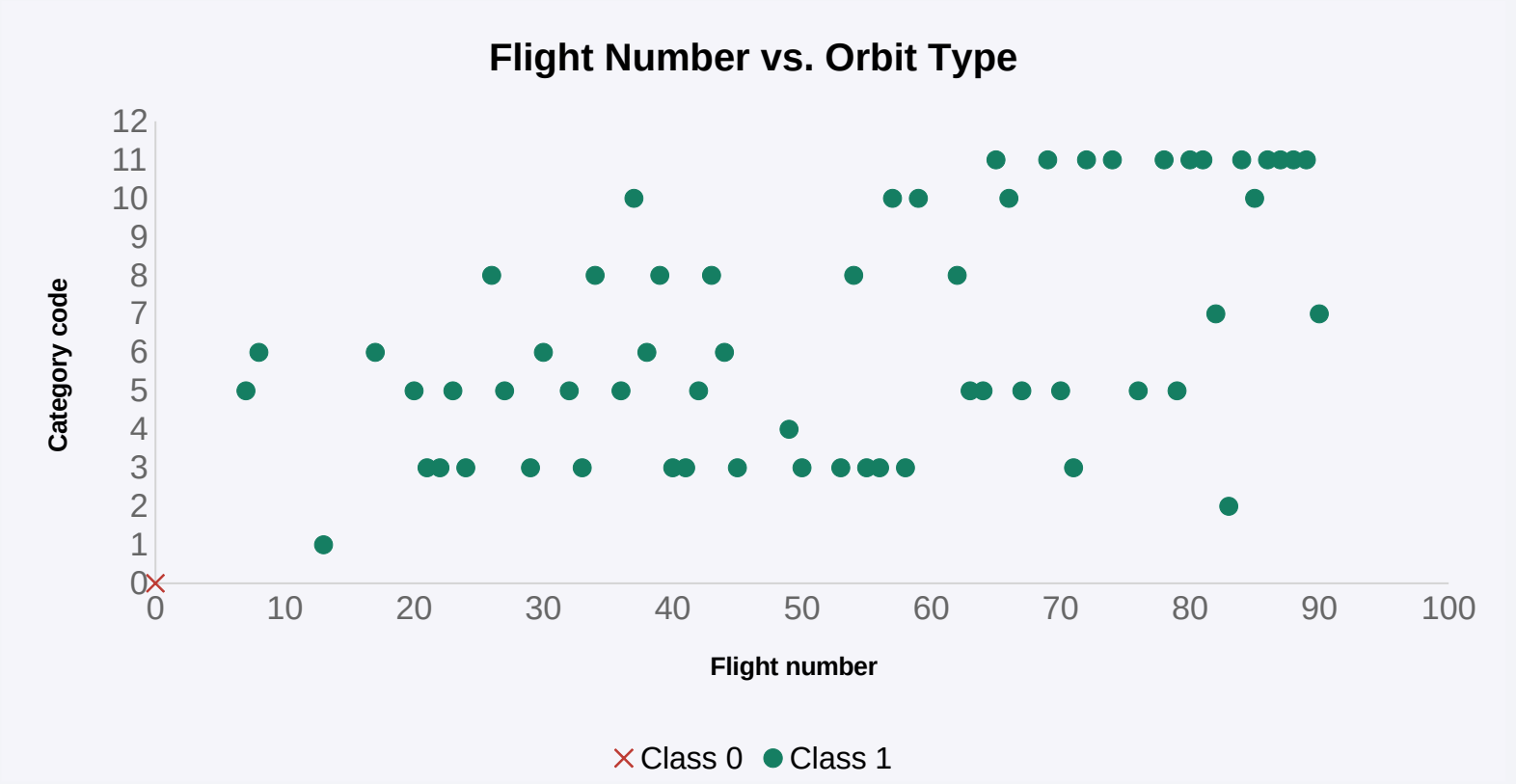


Descriptive rates. Small groups cannot establish an orbit advantage.

Flight Number vs. Orbit Type

Orbit mix changes over the launch sequence. VLEO appears late in this snapshot.

1 ES-L1
2 GEO
3 GTO
4 HEO
5 ISS
6 LEO
7 MEO
8 PO
9 SO
10 SSO
11 VLEO

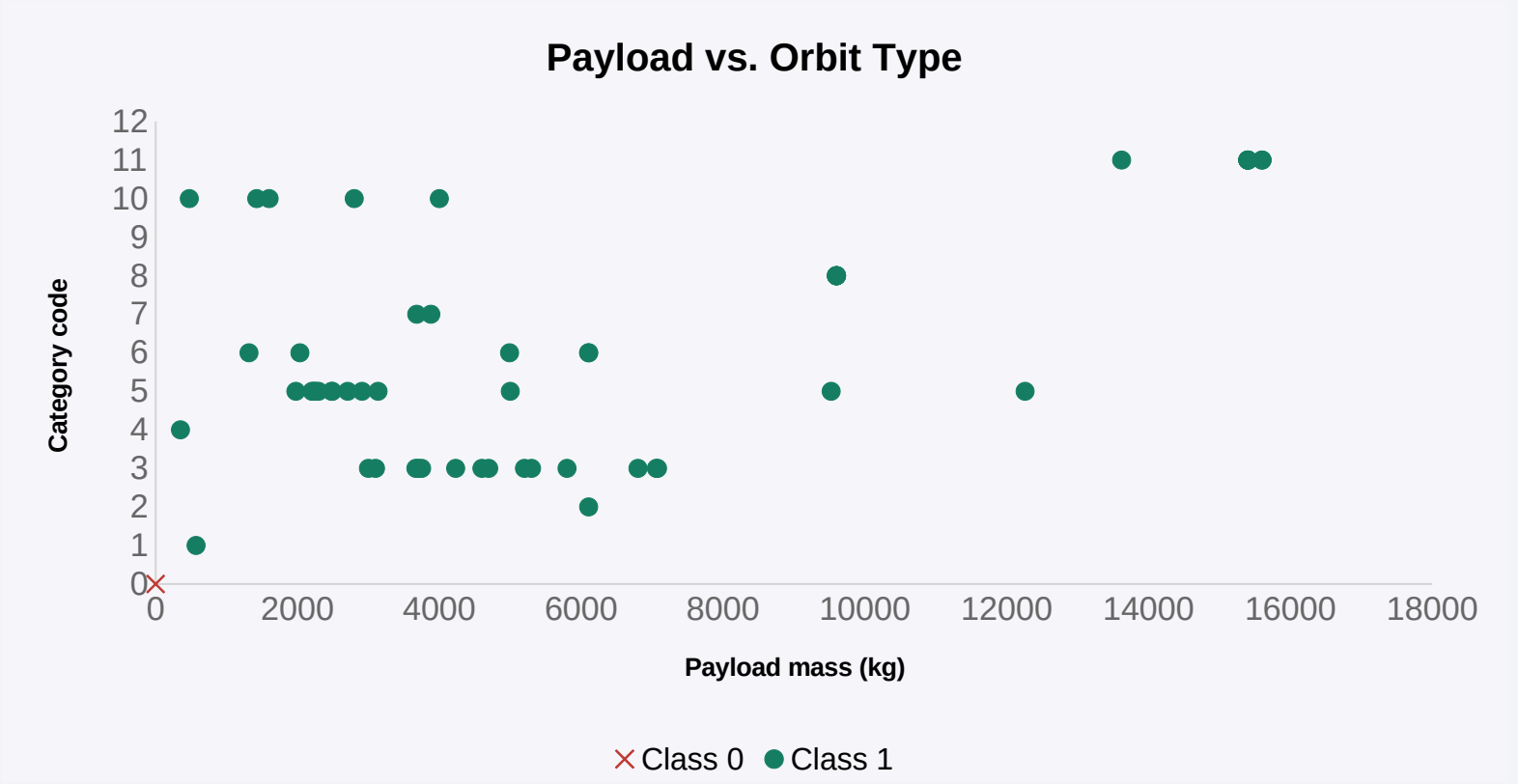


IBM EDA snapshot: 90 launches. Class 1 includes five successful ocean outcomes.

Payload vs. Orbit Type

Heavy VLEO launches often succeed. This association is not causal.

1 ES-L1
2 GEO
3 GTO
4 HEO
5 ISS
6 LEO
7 MEO
8 PO
9 SO
10 SSO
11 VLEO

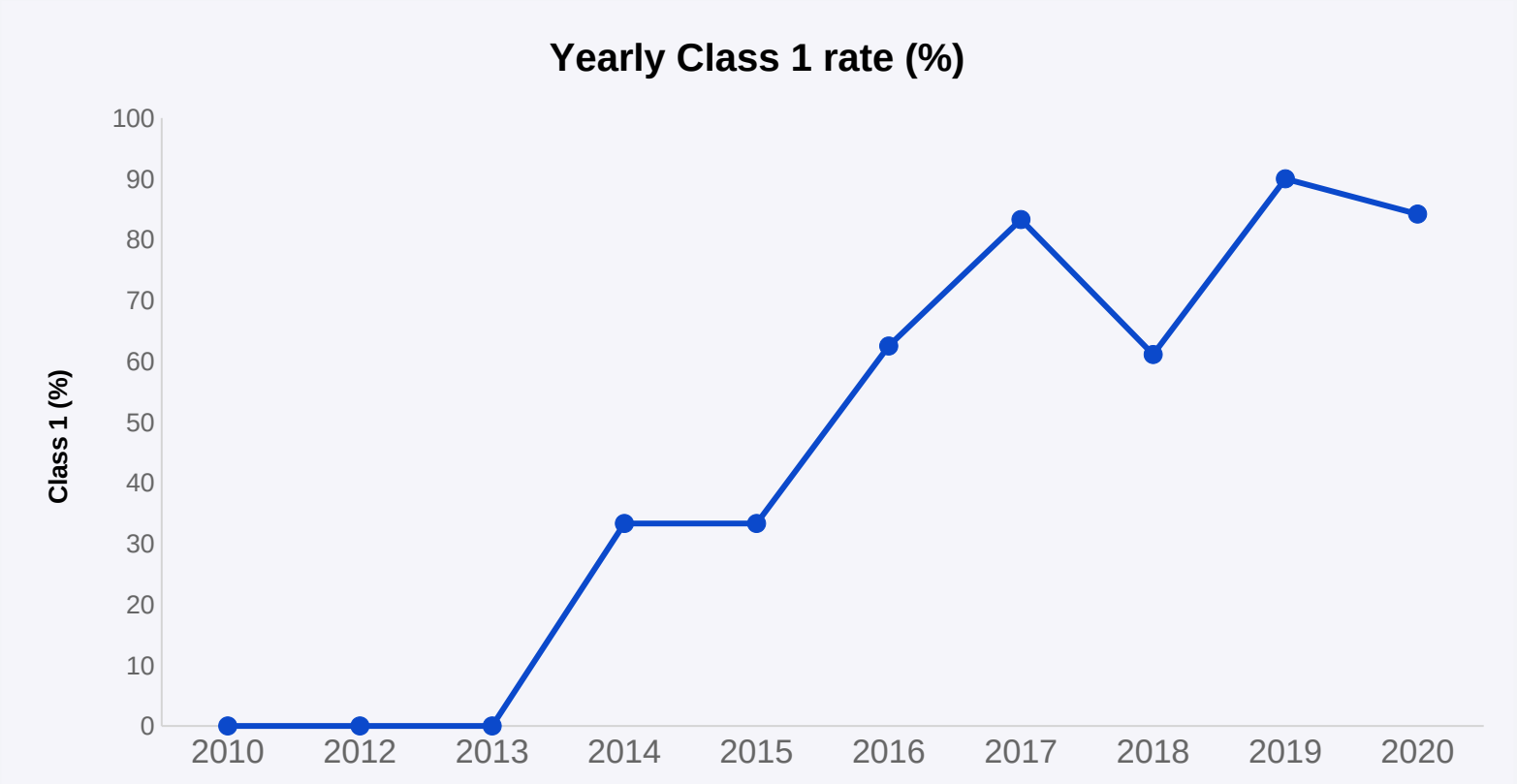


IBM EDA snapshot: 90 launches. Class 1 includes five successful ocean outcomes.

Launch Success Yearly Trend

2017: 15/18 (83.3%)
2018: 11/18 (61.1%)
2019: 9/10 (90.0%)
2020: 16/19 (84.2%)

The trend improves over the early years but is not monotonic.



2011 has no records. 2020 ends on 5 November and is not a full-year census.

All Launch Site Names

Four labels appear. CCAFS LC-40 and SLC-40 refer to the same pad designation across records.

Launch_Site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

```
SELECT DISTINCT Launch_Site
FROM launches ORDER BY Launch_Site
```

Launch Site Names Begin with 'CCA'

First five matches in date order, with a deterministic ORDER BY.

Date	Booster_Version	Launch_Site
2010-06-04	F9 v1.0 B0003	CCAFS LC-40
2010-12-08	F9 v1.0 B0004	CCAFS LC-40
2012-05-22	F9 v1.0 B0005	CCAFS LC-40
2012-10-08	F9 v1.0 B0006	CCAFS LC-40
2013-03-01	F9 v1.0 B0007	CCAFS LC-40

```
SELECT Date,Booster_Version,Launch_Site
FROM launches
WHERE Launch_Site LIKE 'CCA%' ORDER BY Date LIMIT 5
```

Total Payload Mass: NASA (CRS)

Exact customer match: NASA (CRS). This total excludes other NASA customer labels.

kg
45596

```
SELECT SUM(PAYLOAD_MASS__KG_) AS kg
FROM launches
WHERE Customer='NASA (CRS)'
```

Average Payload Mass: F9 v1.1

Prefix match includes all F9 v1.1 booster identifiers.

kg
2534.67

```
SELECT AVG(PAYLOAD_MASS__KG_) AS kg
FROM launches
WHERE Booster_Version LIKE 'F9 v1.1%'
```

First Successful Ground Landing

Minimum date where Landing_Outcome is exactly Success (ground pad).

Date
2015-12-22

```
SELECT MIN(Date) AS Date
FROM launches
WHERE Landing_Outcome='Success (ground pad)'
```

Drone-Ship Successes: 4,000–6,000 kg

Strict inequalities exclude payloads equal to 4,000 or 6,000 kg.

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

```
SELECT DISTINCT Booster_Version
FROM launches
WHERE Landing_Outcome='Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000
```

Mission Outcomes

After trimming spaces: 99 Success, 1 payload-status-unclear success, 1 in-flight failure.

Mission_Outcome	n
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

```
SELECT TRIM(Mission_Outcome) AS Mission_Outcome, COUNT(*) AS n
FROM launches GROUP BY TRIM(Mission_Outcome)
```

Boosters with Maximum Payload

All 12 tied records carry 15,600 kg. No arbitrary single winner.

Booster version	Booster version	Payload (kg)
F9 B5 B1048.4	F9 B5 B1049.5	15,600
F9 B5 B1049.4	F9 B5 B1060.2	15,600
F9 B5 B1051.3	F9 B5 B1058.3	15,600
F9 B5 B1056.4	F9 B5 B1051.6	15,600
F9 B5 B1048.5	F9 B5 B1060.3	15,600
F9 B5 B1051.4	F9 B5 B1049.7	15,600

```
SELECT Booster_Version,PAYLOAD_MASS__KG_
FROM launches
WHERE PAYLOAD_MASS__KG_=(SELECT MAX(PAYLOAD_MASS__KG_)
FROM launches)
```

2015 Drone-Ship Landing Failures

Two failures match the exact drone-ship label. Other failure categories are excluded.

Date	Booster_Version	Launch_Site
2015-01-10	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	F9 v1.1 B1015	CCAFS LC-40

```
SELECT Date,Booster_Version,Launch_Site
FROM launches
WHERE Date LIKE '2015%' AND Landing_Outcome='Failure (drone ship)'
```

Landing Outcomes: 2010–2017

Inclusive date range: 4 June 2010 through 20 March 2017. 31 records in total.

Landing_Outcome	n
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

```
SELECT Landing_Outcome,COUNT(*) AS n
FROM launches
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY n DESC
```



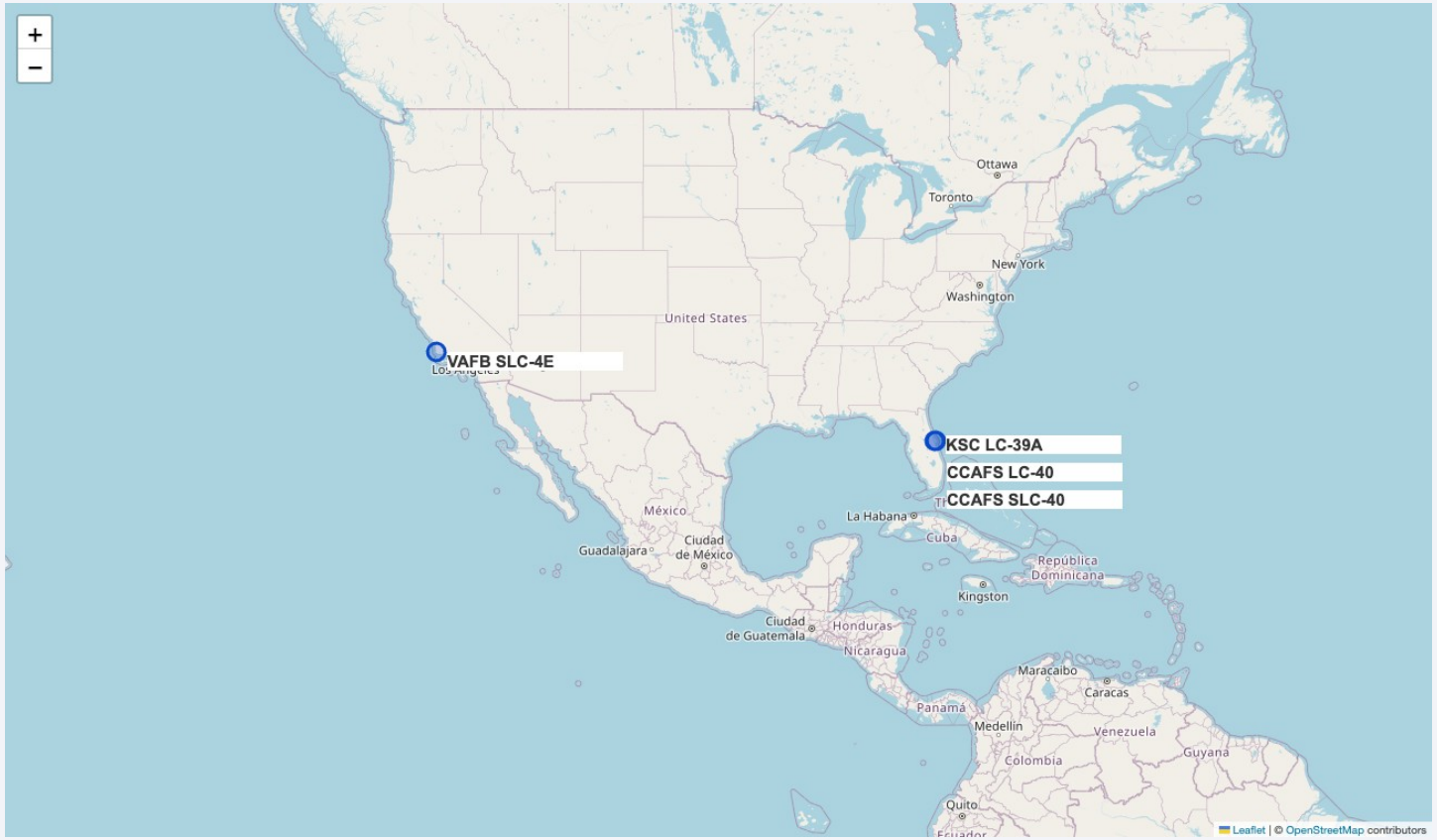
Section 3

Launch Sites Proximities Analysis

Launch Sites in the Course Snapshot

Four site labels span Florida and California.

The two CCAFS labels nearly overlap. Keep raw labels visible when auditing the source.

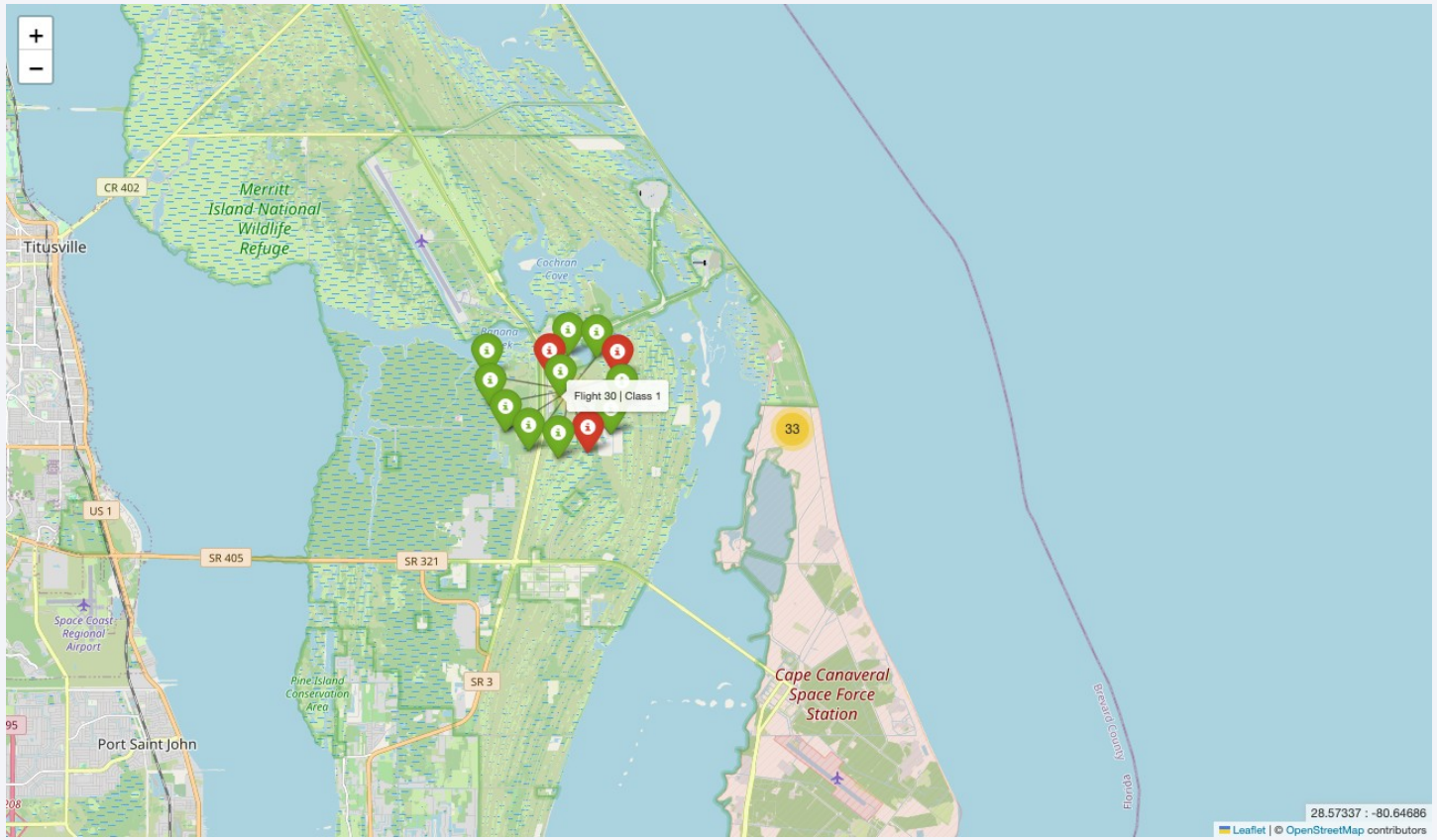


Reconstructed Folium view. Basemap: OpenStreetMap contributors. Source: IBM geographic snapshot.

Launch Outcomes at Cape Canaveral

The Folium cluster holds all 56 records.

Expand a cluster to inspect green Class 1 and red Class 0 markers at the same coordinates.

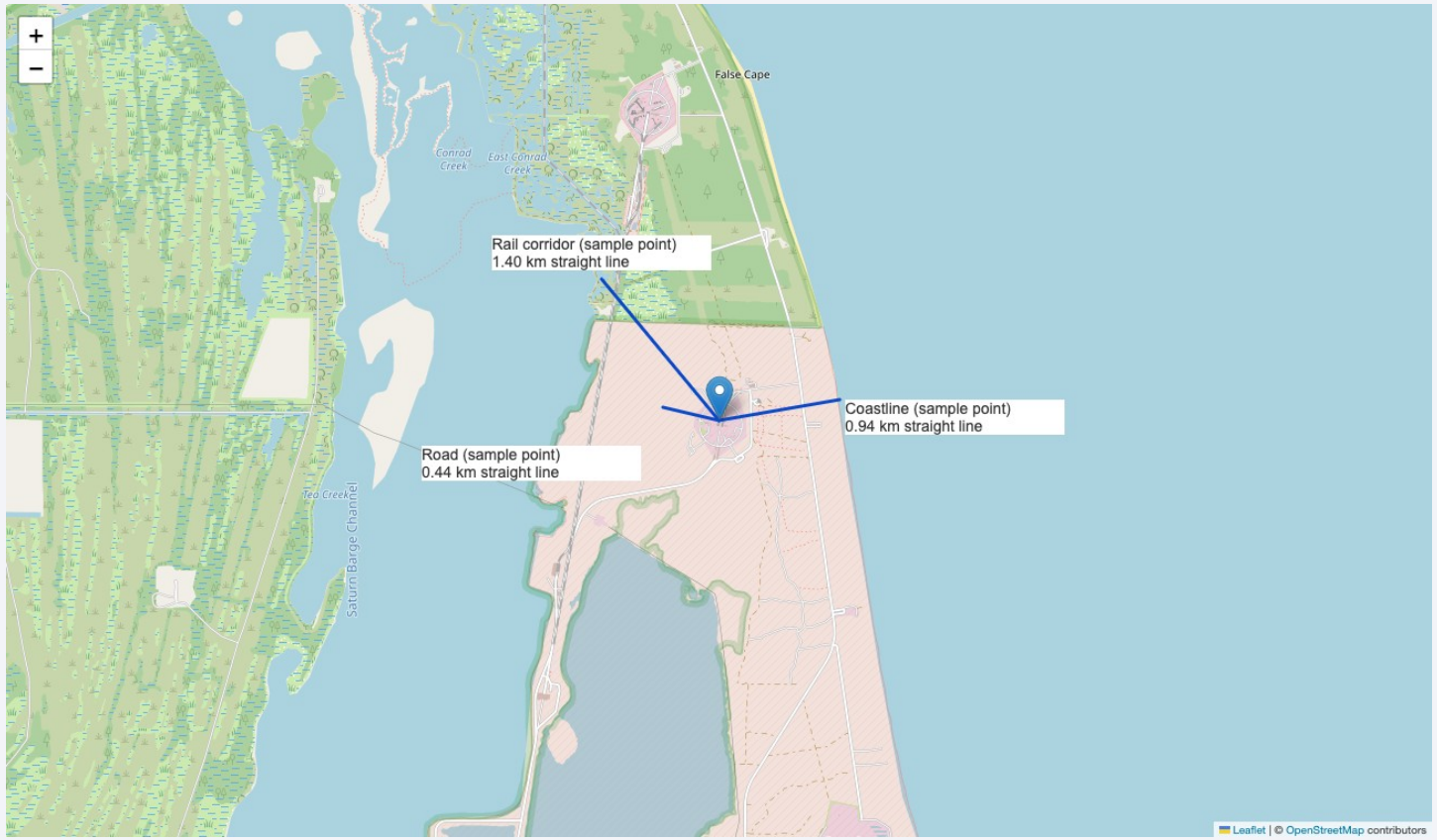


Reconstructed Folium view. Basemap: OpenStreetMap contributors. Source: IBM geographic snapshot.

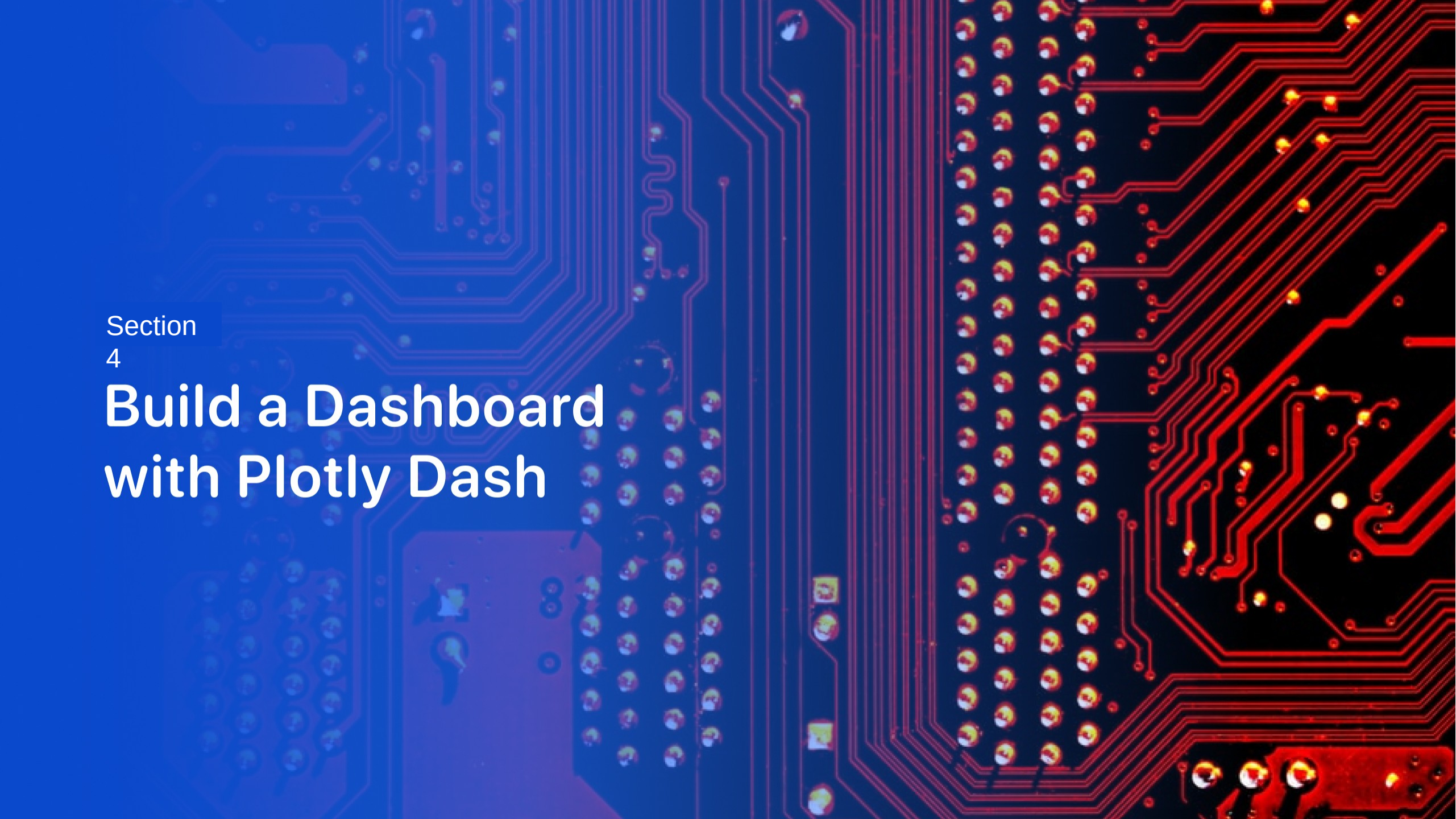
CCAFS LC-40 Proximity Example

Selected points:
Coastline: 0.94 km
Road: 0.44 km
Rail corridor: 1.40 km

Approximate sample points. Distances are straight line, not nearest-feature measurements.



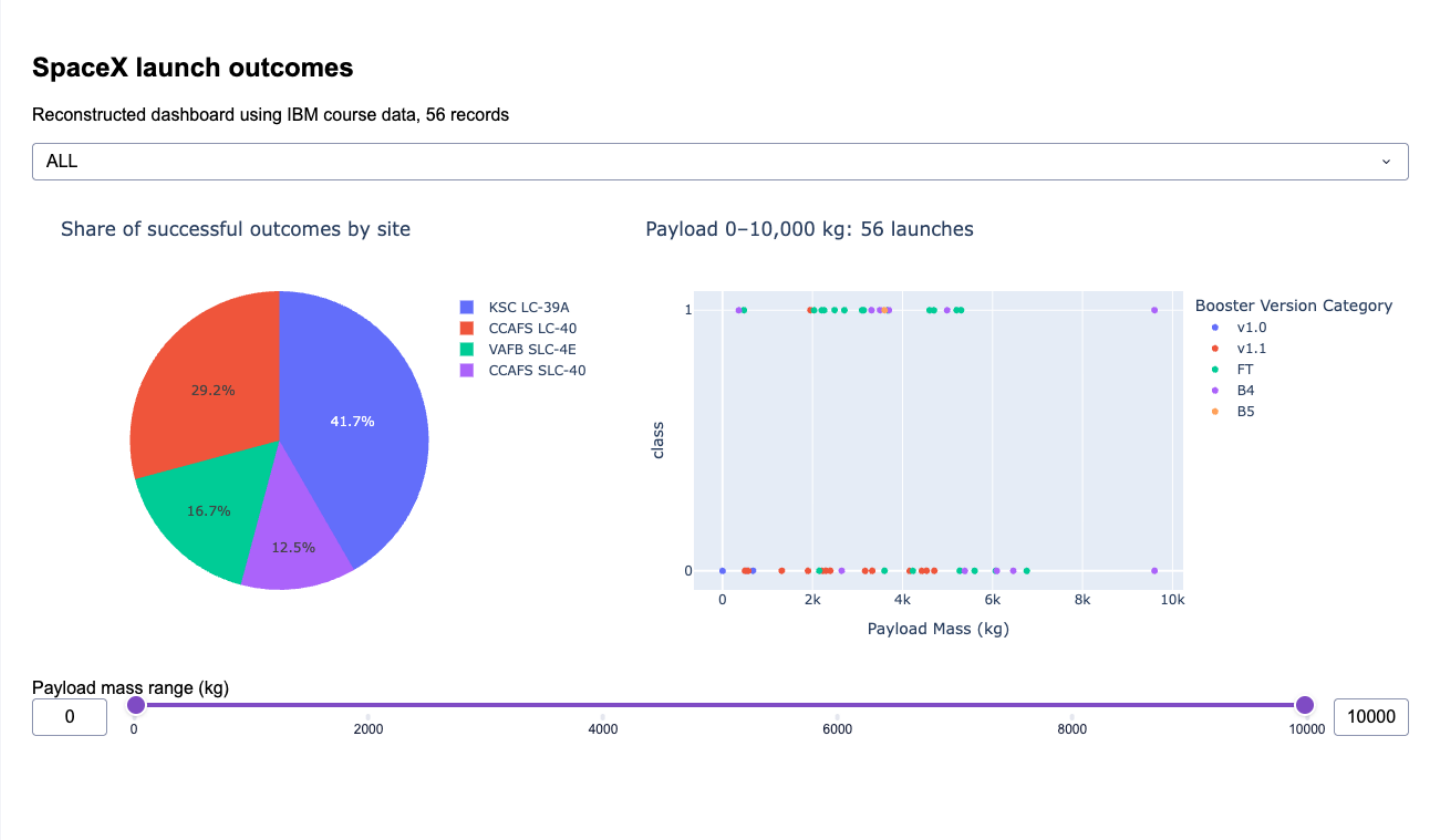
Reconstructed Folium view. Basemap: OpenStreetMap contributors. Source: IBM geographic snapshot.


Section
4
Build a Dashboard
with Plotly Dash

Dashboard: Successful Outcomes by Site

KSC: 10 of 24 successes (41.7%).

This pie compares success counts across sites. It does not show each site's success probability.



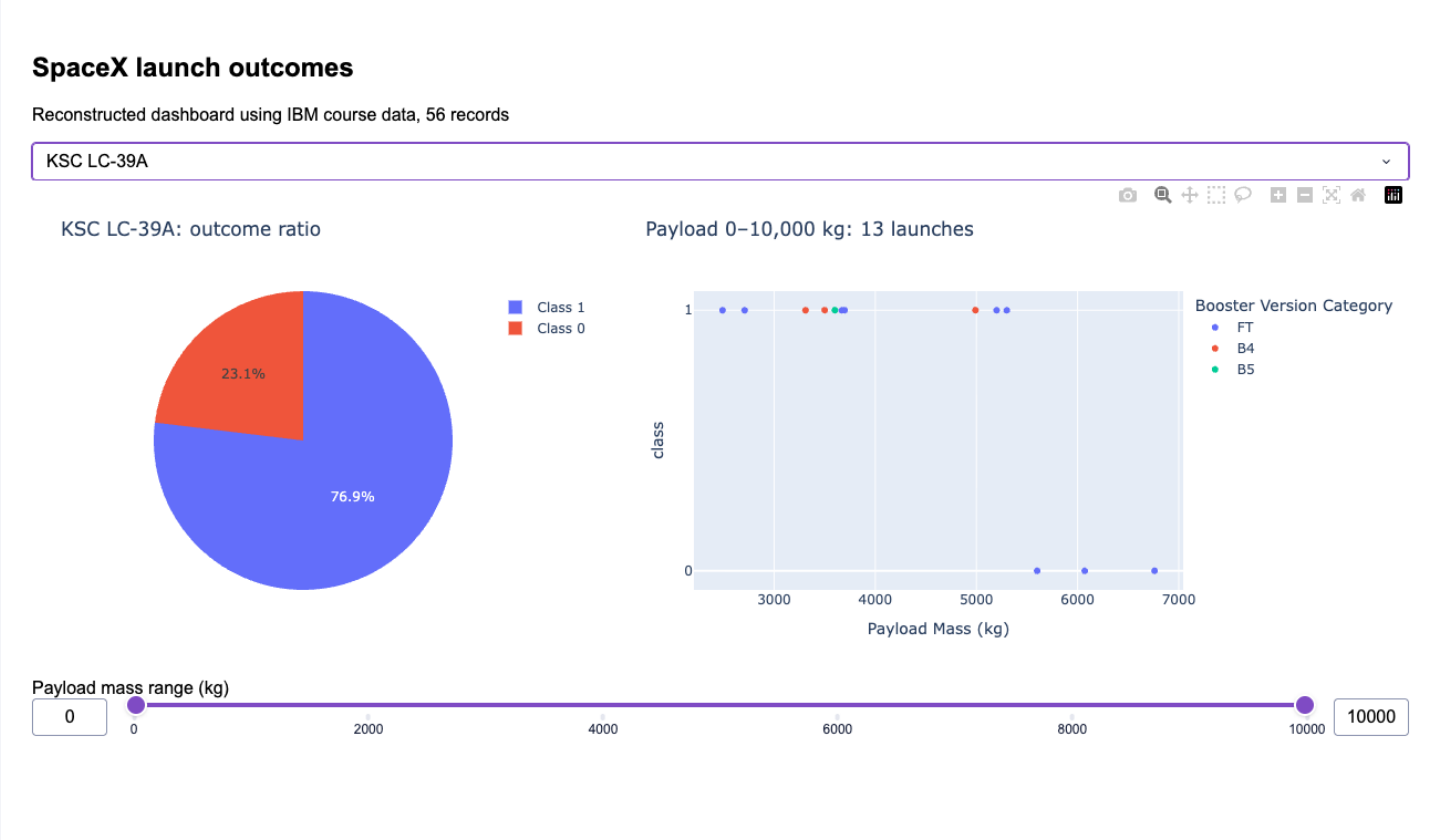
Reconstructed, functioning Plotly Dash application. IBM dashboard snapshot: 56 launches.

Dashboard: KSC LC-39A Outcome Ratio

KSC has the highest site ratio in this snapshot:

10 Class 1 / 13 launches = 76.9%.

The sample contains only 13 launches.



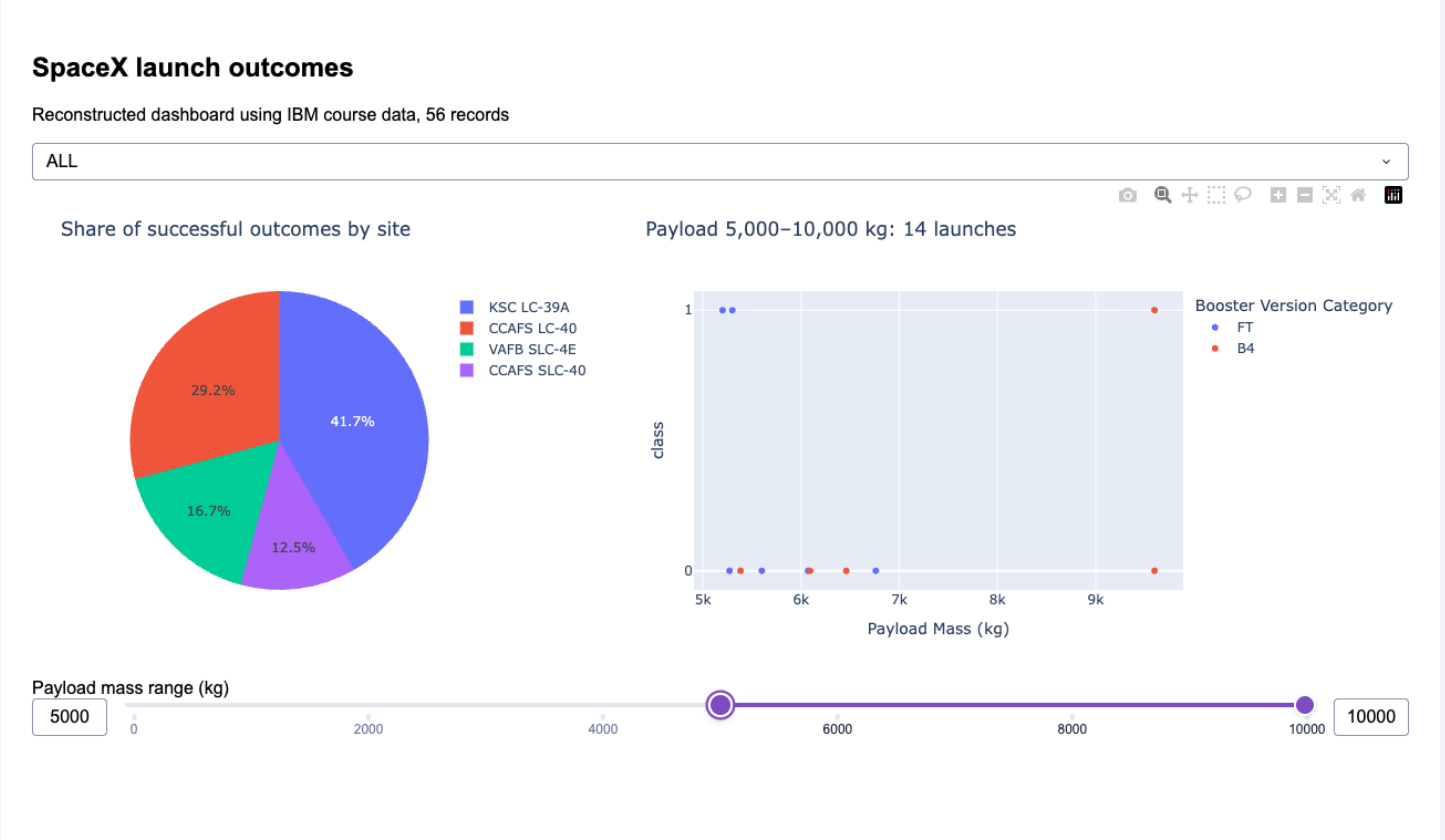
Reconstructed, functioning Plotly Dash application. IBM dashboard snapshot: 56 launches.

Dashboard: Payload Range Filter

The 5,000–10,000 kg range contains 14 launches and 5 Class 1 outcomes (35.7%).

Compare this selected range with the full range on slide 39.

B5 has 1/1 success overall. That single launch cannot establish version superiority.



Reconstructed, functioning Plotly Dash application. IBM dashboard snapshot: 56 launches.

Section
5

Predictive Analysis (Classification)

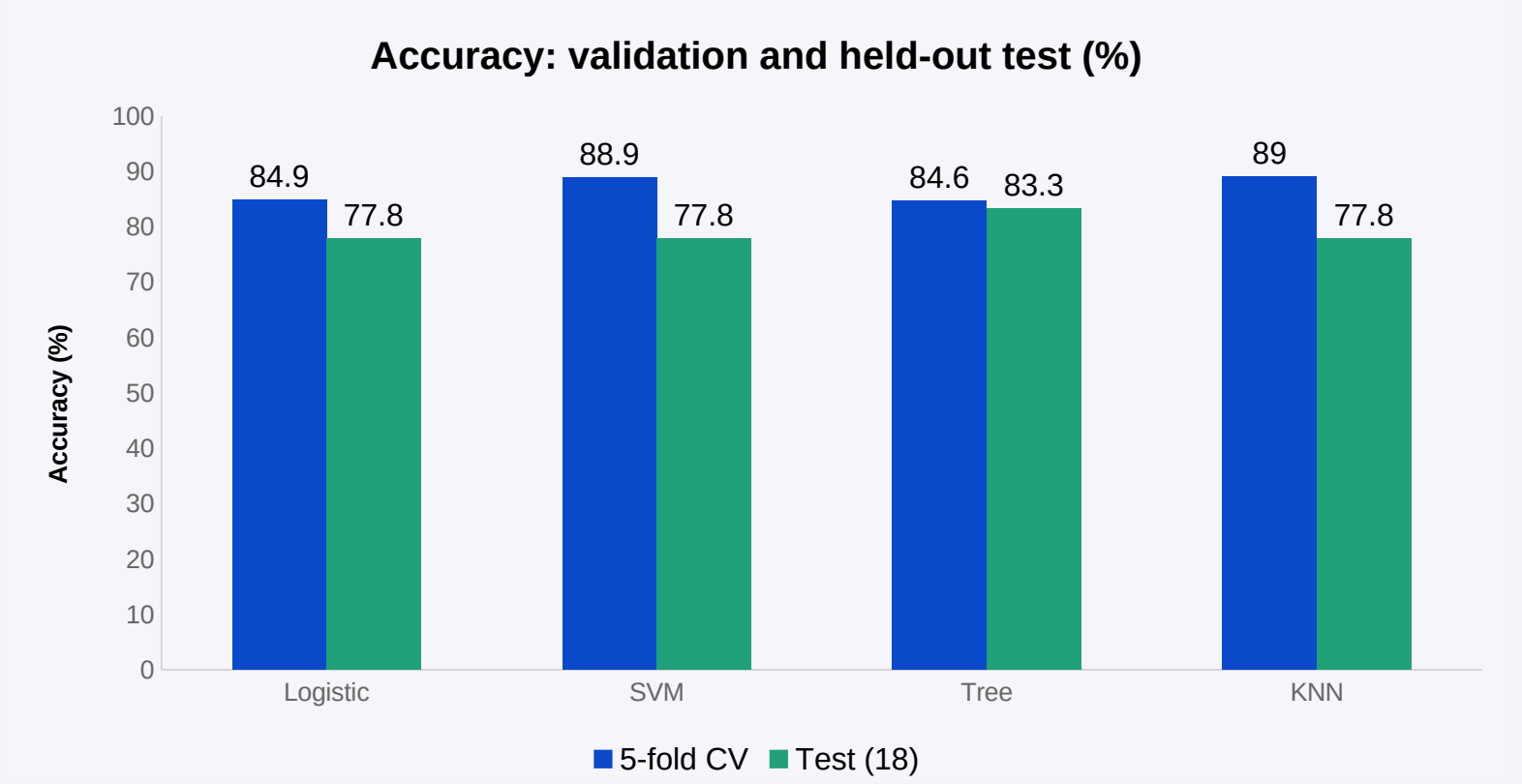


Classification Accuracy

Tree has the highest observed test score: 15/18.

KNN leads CV: 89.0%, versus SVM 88.9%.

Always-Class-1 test baseline: 12/18 (66.7%).



Model selection uses training CV. Test-set ranking is descriptive and must not guide retuning.

Confusion Matrices

Decision tree	Predicted 0	Predicted 1
Actual 0	3	3
Actual 1	0	12

KNN (CV selected)	Predicted 0	Predicted 1
Actual 0	3	3
Actual 1	1	11

Decision tree: accuracy 83.3%, precision 80.0%, recall 100%, F1 88.9%.

KNN: accuracy 77.8%, precision 78.6%, recall 91.7%, F1 84.6%.

Both falsely predict success for 3 of the 6 actual Class 0 launches.

Conclusions

Historical launch outcomes improve across the early flight sequence. Orbit, payload and site remain entangled with time and mission mix.

The course target includes ocean successes and groups unknown outcomes with failures. It does not directly measure booster recovery for reuse.

The tree achieves 83.3% test accuracy, but the test set has only 18 launches. KNN and SVM are effectively tied in CV.

Next steps: validate prelaunch feature availability, use time-based testing, refine the recovery target and quantify uncertainty.

Appendix: Reproducibility

EDA / ML: 90 rows, 60 Class 1 and 30 Class 0. Train/test split: 72/18, stratified, random_state=42.

Predictors: FlightNumber, PayloadMass, Flights, Block, ReusedCount, Orbit, LaunchSite, GridFins, Reused and Legs.

Exclude Outcome, Class, LandingPad, Serial, Date and redundant location fields from X. ReusedCount needs a prelaunch-availability audit.

Five CV folds, shuffled with seed 42. scikit-learn 1.9.1. Executed notebooks and Python files are available in the linked repository.

Completed notebooks and Python files:
https://github.com/mohitpingale24-alt/IBM-Stock-Data-Analysis/tree/main/spacex-capstone

Appendix: Hyperparameter Search

Model	Grid	Selected
Logistic	C: .01, .1, 1, 10, 100	C=.1
SVM	C: .1, 1, 10, 100 kernel: linear/rbf; gamma: scale/.01/.1	C=1, rbf, gamma=scale
Tree	depth: 2,3,4,5,None; leaf: 1,2,4 criterion: gini/entropy	depth=3, leaf=4, gini
KNN	neighbors: 3,5,7,9; p: 1,2 weights: uniform/distance	k=3, p=1, distance

Best mean CV accuracy selects each configuration. No test data enter preprocessing or tuning.

Appendix: Evidence and Limitations

Completed here: CSV audits, ten SQLite queries, six EDA charts, four trained classifiers, Folium maps and working Dash views.

API: official IBM snapshot, with an optional live collector. Web scraping: executed independently on archived Wikipedia HTML. Modeling results continue to use the IBM snapshot.

Maps preserve course coordinates. Proximity points are approximate demonstrations. No surveyed distance or site-safety claim follows.

Random splitting can share booster history across sets. Prior payload imputation and possible retrospective fields limit leakage guarantees.

Appendix: Sources

IBM course data and template (retrieved 27 October 2026)
https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/

EDA / ML: datasets/dataset_part_1.csv and dataset_part_2.csv
SQL: labs/module_2/data/Spacex.csv
Maps: datasets/spacex_launch_geo.csv
Dashboard: datasets/spacex_launch_dash.csv
Template: labs/module_5/ds-capstone-template-coursera.pptx

API methodology: github.com/r-spacex/SpaceX-API
Scraping methodology: Wikipedia, List of Falcon 9 and Falcon Heavy launches
Basemap attribution: OpenStreetMap contributors

Completed notebooks and Python files:
https://github.com/mohitpingale24-alt/IBM-Stock-Data-Analysis/tree/main/spacex-capstone

Thank you!



SpaceX Falcon 9 historical analysis